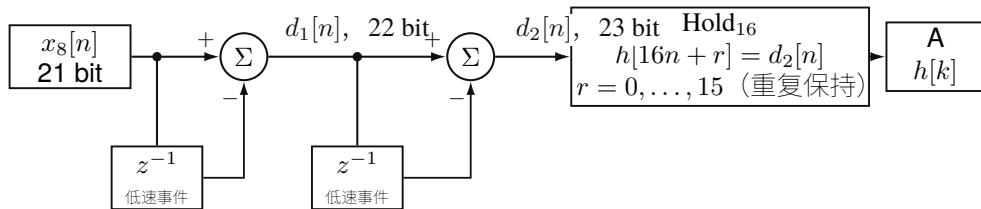


16 倍三级 CIC 插值器的算法级等效信流图

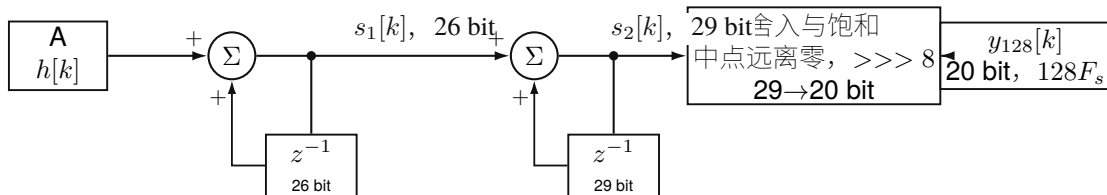
低速 $C^3 \rightarrow \uparrow 16 \rightarrow I^3 \equiv$ 低速 $C^2 \rightarrow \text{Hold}_{16} \rightarrow I^2$

(a) 低速二阶差分与 16 倍保持



$$\begin{aligned} d_1[n] &= x_8[n] - x_8[n-1] \\ d_2[n] &= d_1[n] - d_1[n-1] \end{aligned}$$

(b) 高速二阶积分与定点输出



$$\begin{aligned} s_1[k] &= s_1[k-1] + h[k] \\ s_2[k] &= s_2[k-1] + s_1[k] \end{aligned}$$

z_ℓ^{-1}/z_h^{-1} 为事件延时;
全链为同一 $128F_s$ 主时钟。